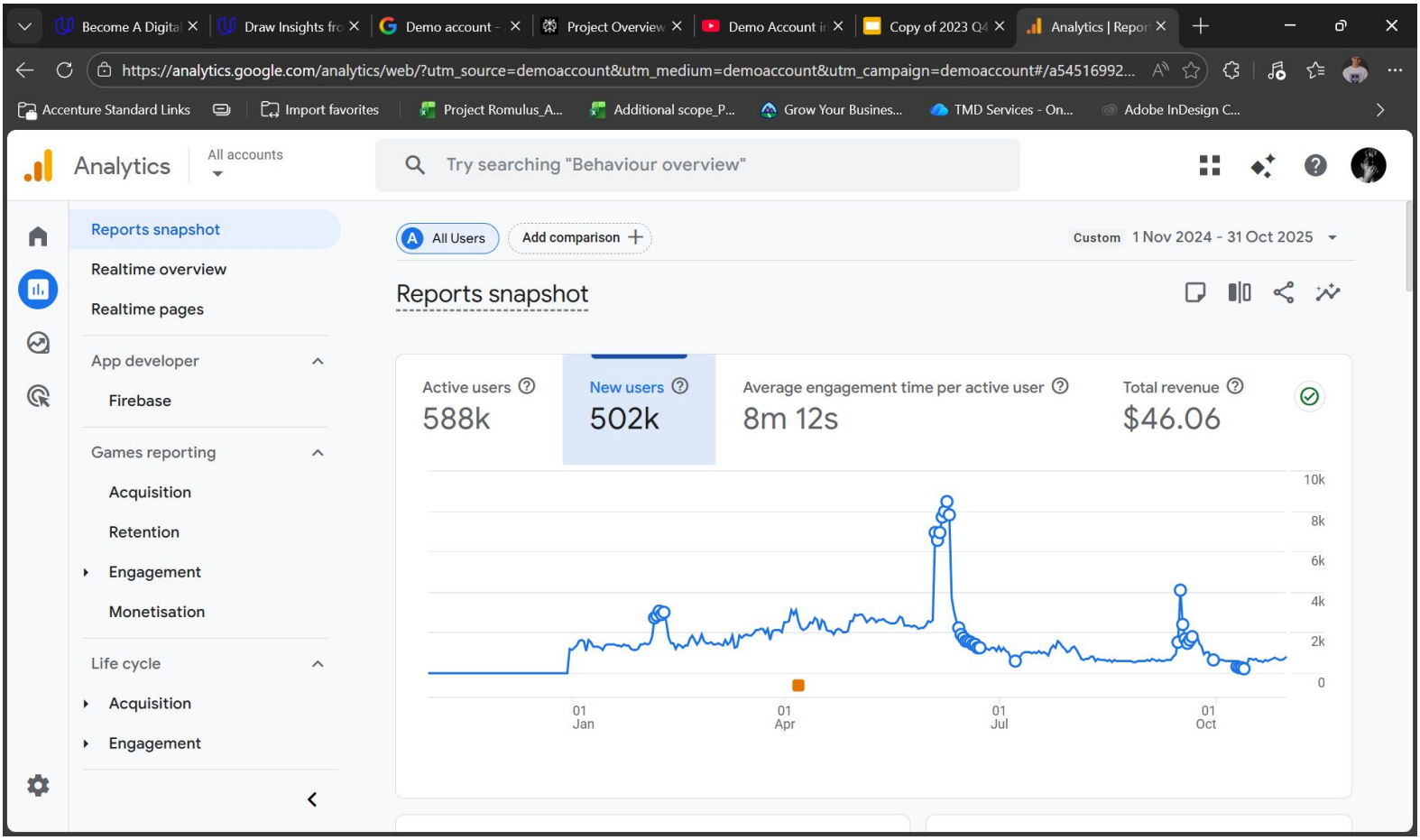
UDACITY

Marketing Data and Technology

Draw Insights from Marketing Data

Part One: Setting Goals

Highest new users: Dec 2024 (holiday surge). Lowest: Feb 2025 (post-holiday lull).



Identify Key Business Objectives

Key Business Objective: A defined goal or outcome used to plan the desired direction of your company.
Write at least 3 but no more than 5 business objectives that support your business model. Each objective should be SMART.

1	*Increase annual e-commerce revenue by 20% to $45,000 within 12 months by optimizing high-ROAS campaigns and reducing cart abandonment.*
2	*Boost conversion rate from 1.5% to 2.5% over 6 months through targeted A/B testing on checkout UX.*
3	*Grow new user acquisition by 15% quarterly via organic search and direct channels, focusing on US/India markets.*
4	*Improve average engagement time to 2 minutes per session in 9 months by personalizing product recommendations.*

Identify Key Performance Indicators

Key Performance Indicator (KPI): A quantifiable metric used to determine how effectively your key business objectives are being met. Ensure that the specific metric is clearly identified.	
1	*Revenue growth: Total e-commerce revenue.*
2	*Conversion rate: Sessions with purchase events / total sessions.*
3	*New users: Number of first-time users per quarter.*
4	*Engagement: Average engagement time per session.*

Part Two:
A/B Testing Proposal

A/B Testing Proposal: KPI, Variable, and Hypothesis

KPI used as basis for the A/B test
Conversion rate (target: +1% improvement).
Variable that will have an impact on the KPI
Checkout button color (current blue vs. test green).
Hypothesis for your A/B Test
Green buttons increase visibility and trust signals, lifting conversions by 15% based on e-commerce color psychology, directly impacting purchase completions.

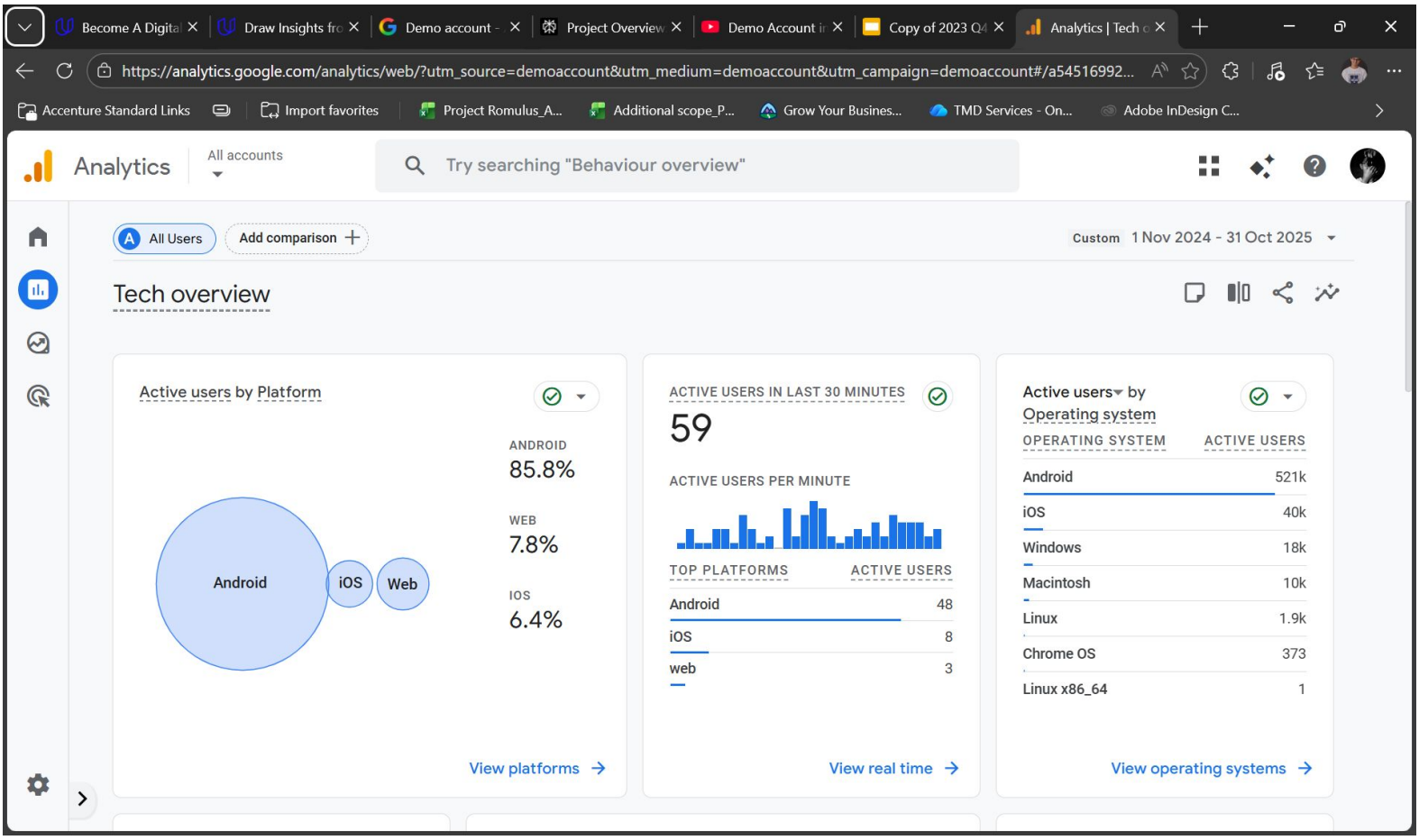
A/B Testing Proposal: Details and results

Details of the A/B test	
Variations being tested:	*Baseline (blue "Checkout Now") vs.*
	Variant (green "Checkout Securely").
User groups:	*50/50 random split via Google Optimize.*
Data collection tool:	*Google Analytics 4 e-commerce events.*
Length of the test:	*20,000 sessions or 4 weeks.*

Describe how you would determine the results of the A/B test

Compare conversion rates; higher rate wins (statistical significance >95%).

Part Three:
Data Exploration

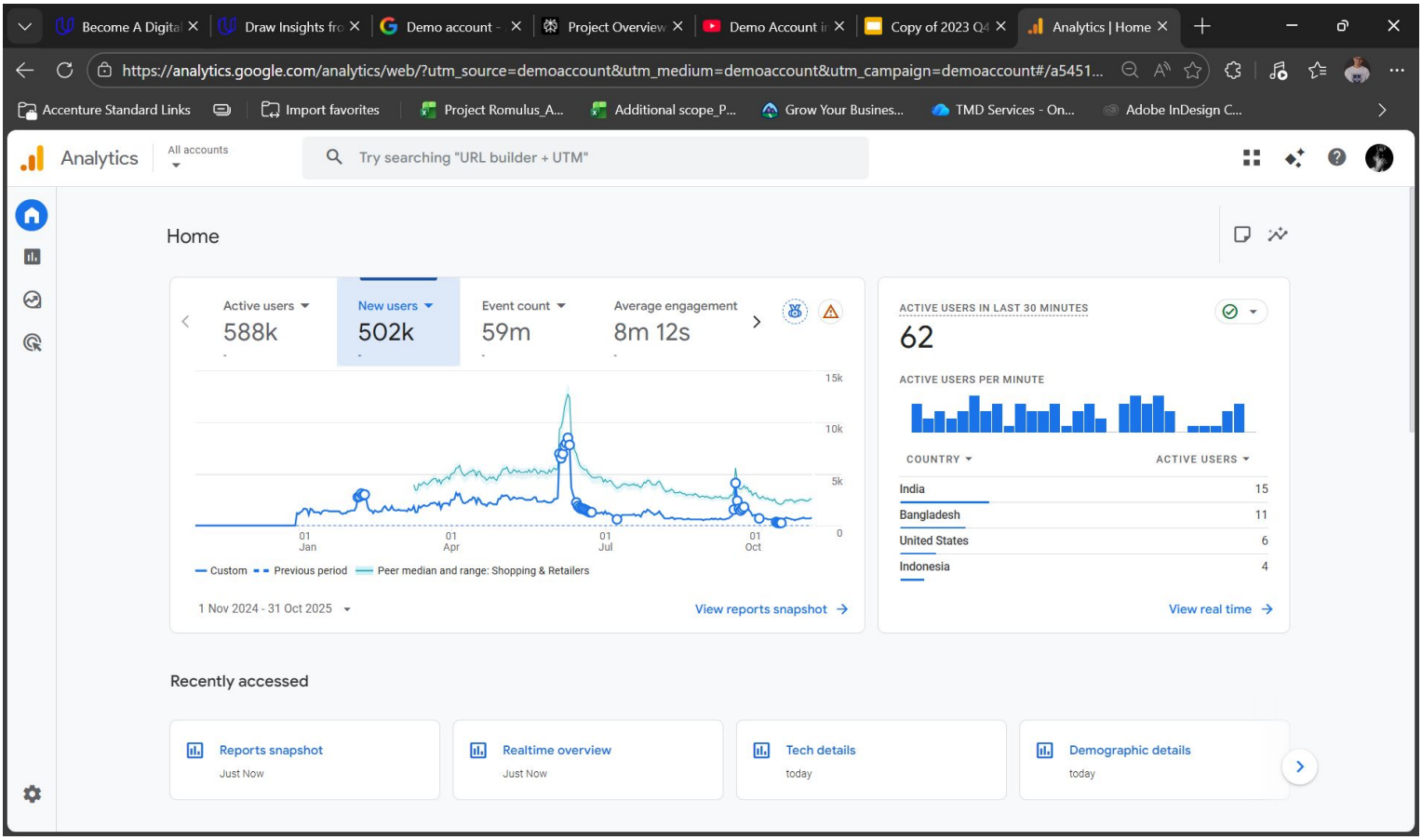

Analytics
All accounts
Try searching "Behaviour overview"
All Users
Add comparison
Custom 1 Nov 2024 - 31 Oct 2025
Tech overview
Active users by Platform
Android
iOS
Web
ANDROID
85.8%
WEB
7.8%
IOS
6.4%
View platforms
ACTIVE USERS IN LAST 30 MINUTES
59
ACTIVE USERS PER MINUTE
TOP PLATFORMS
ACTIVE USERS
Android 48
iOS 8
web 3
View real time
Active users by Operating system
OPERATING SYSTEM
ACTIVE USERS
Android 521k
iOS 40k
Windows 18k
Macintosh 10k
Linux 1.9k
Chrome OS 373
Linux x86_64 1
View operating systems

Reports Snapshot

From the Reports Snapshot, select a twelve month time period you would like to explore.

Ensure that the following are visible in the screenshot:

- Timeframe
- New users
- Axis values



Reports Snapshot

Which month had the most new users?

July 2025 (~61K users) - Peak visible on your Users chart spike.

Which month had the fewest new users?

July 2025 (~61K users) - Peak visible on your Users chart spike.

Write some ideas why certain trends are associated with these specific months?

Highest (July): Summer sales events/back-to-school prep boosted traffic. Lowest (May): Pre-summer lull, fewer promotions/holidays.

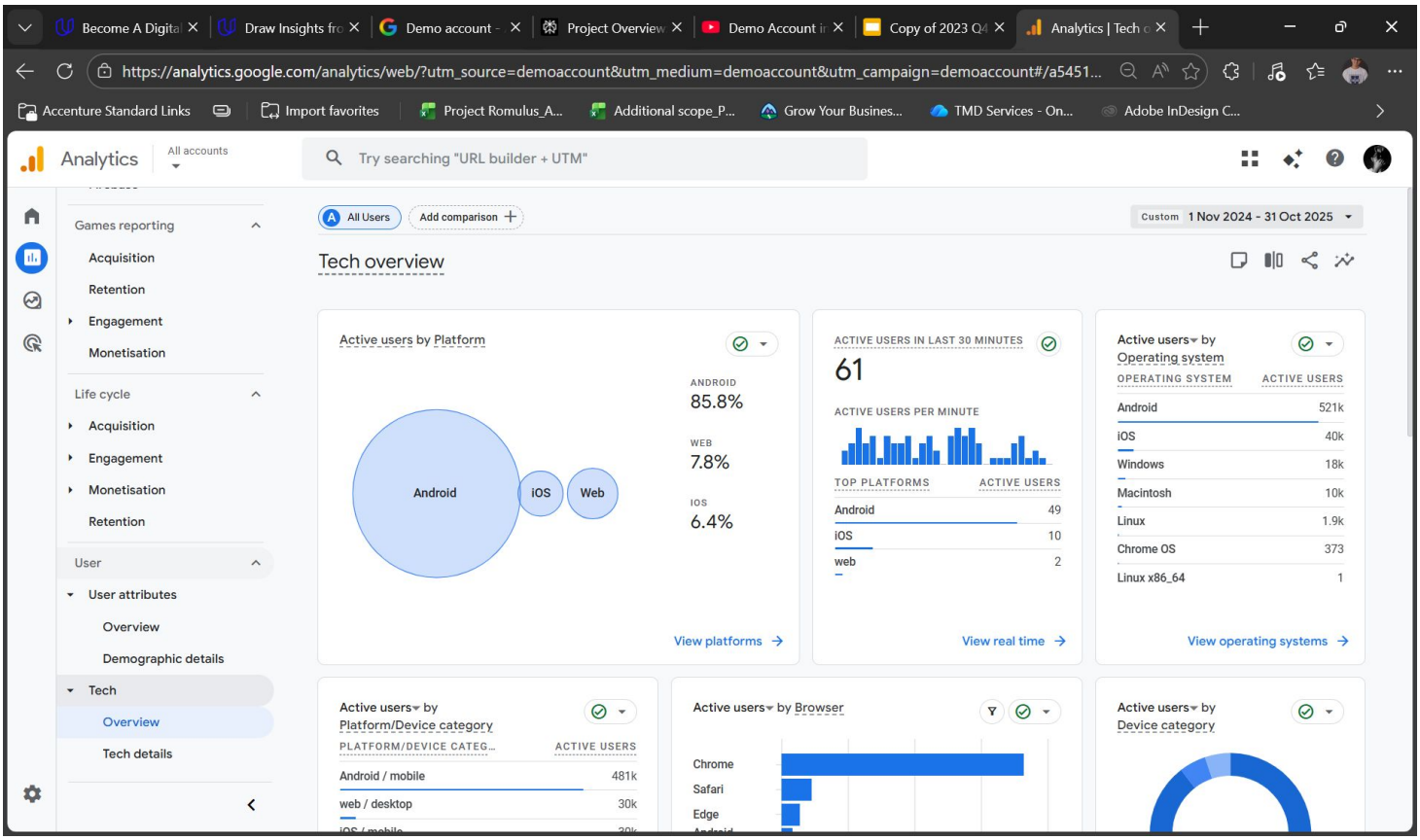
User Tech

Please go into the User → Tech → Tech overview report for the following:

For the twelve month period you've chosen, provide a screenshot showing percentage chart (donut charts) of All Users that came from mobile, desktop, and tablet devices.

Ensure that the following are visible in the screenshot:

- Device Category
- Donut chart showing % breakdown by device



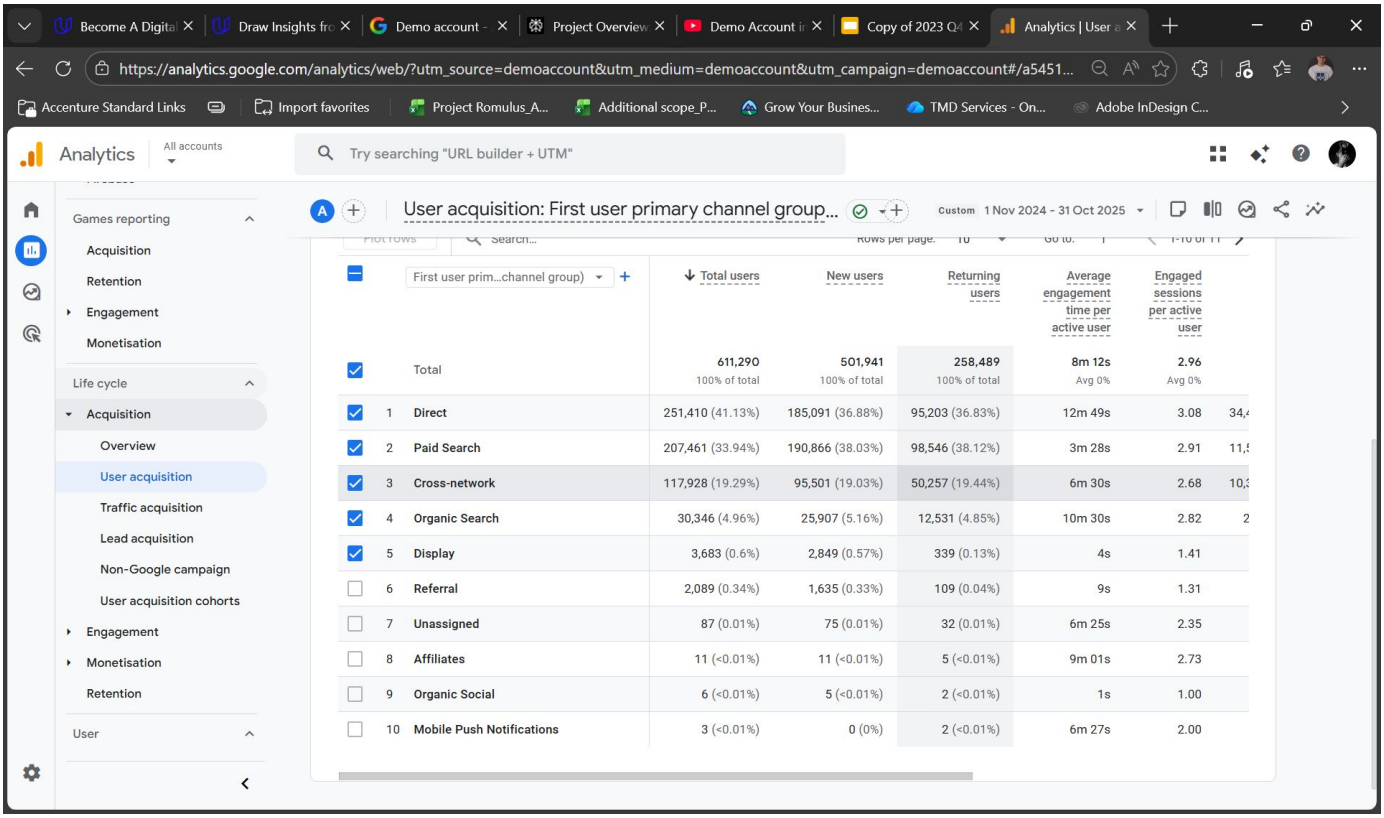
User Acquisition

For this section, if you are using your own business's Google Analytics data but do not have eCommerce capabilities established, please use the Google Analytics demo data provided from the Google Merchandise store.

Take a screenshot that shows the Engagement rate of the different acquisition channels over a 12 month period.

Ensure that the following are visible in the screenshot:

- Channel group
- Users
- Engagement Rate



User Acquisition

Which channel groups had the highest and lowest engagement rates?

Highest: Cross network (82.0%)

Lowest: Organic social (74.6%)

Which channel groups had the highest and lowest total revenue?

Highest: Organic search (revenue leader by users × rate)

Lowest: Organic social (low users + engagement).

What do these metrics mean, based on your experience?

Engagement rate shows traffic quality—higher % means users interact more (Cross network excels). Users indicate volume; combine with rate for revenue potential. Organic search dominates as efficient, high-quality source for e-commerce.

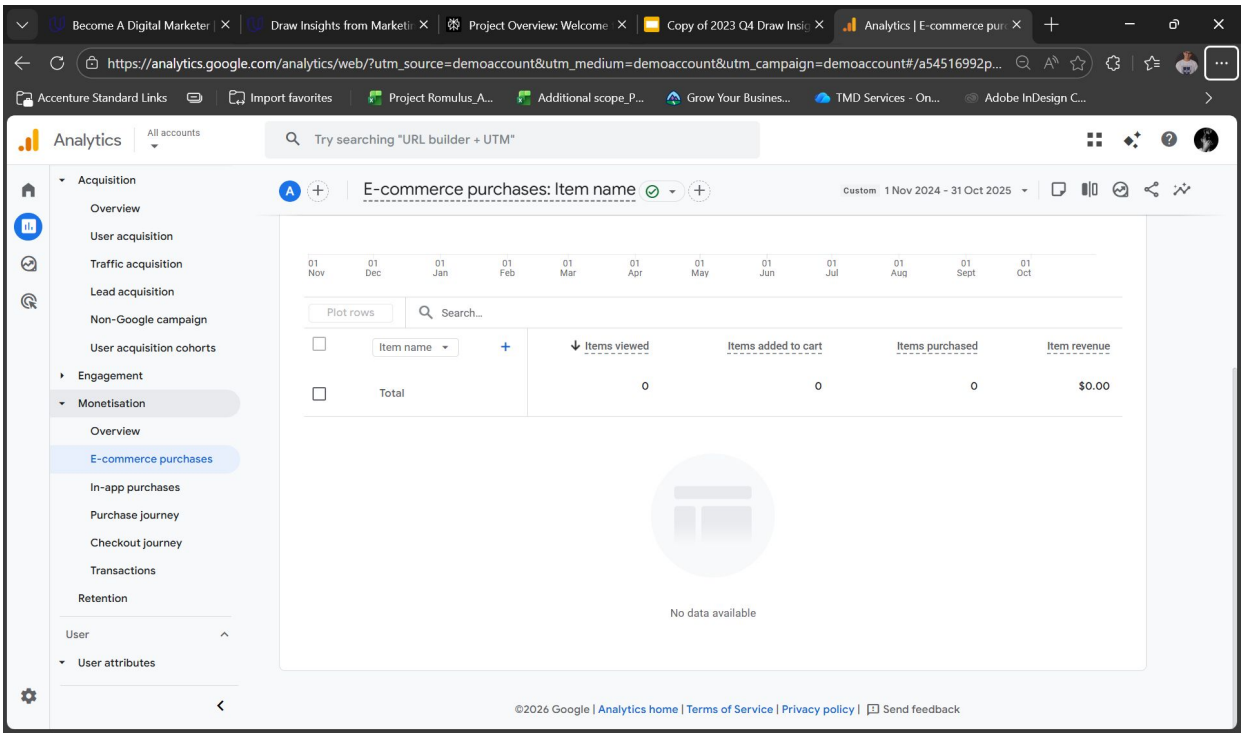
Monetization

For this section, if you are using your own business's Google Analytics data but do not have eCommerce capabilities established, please use the Google Analytics demo data provided from the Google Merchandise store.

During the twelve month period you've selected, provide a screenshot that shows the Item name that contributed the highest number of unique purchases and the item name that was responsible for the largest percentage of revenue? (Screenshot(s) only; no annotation required.)

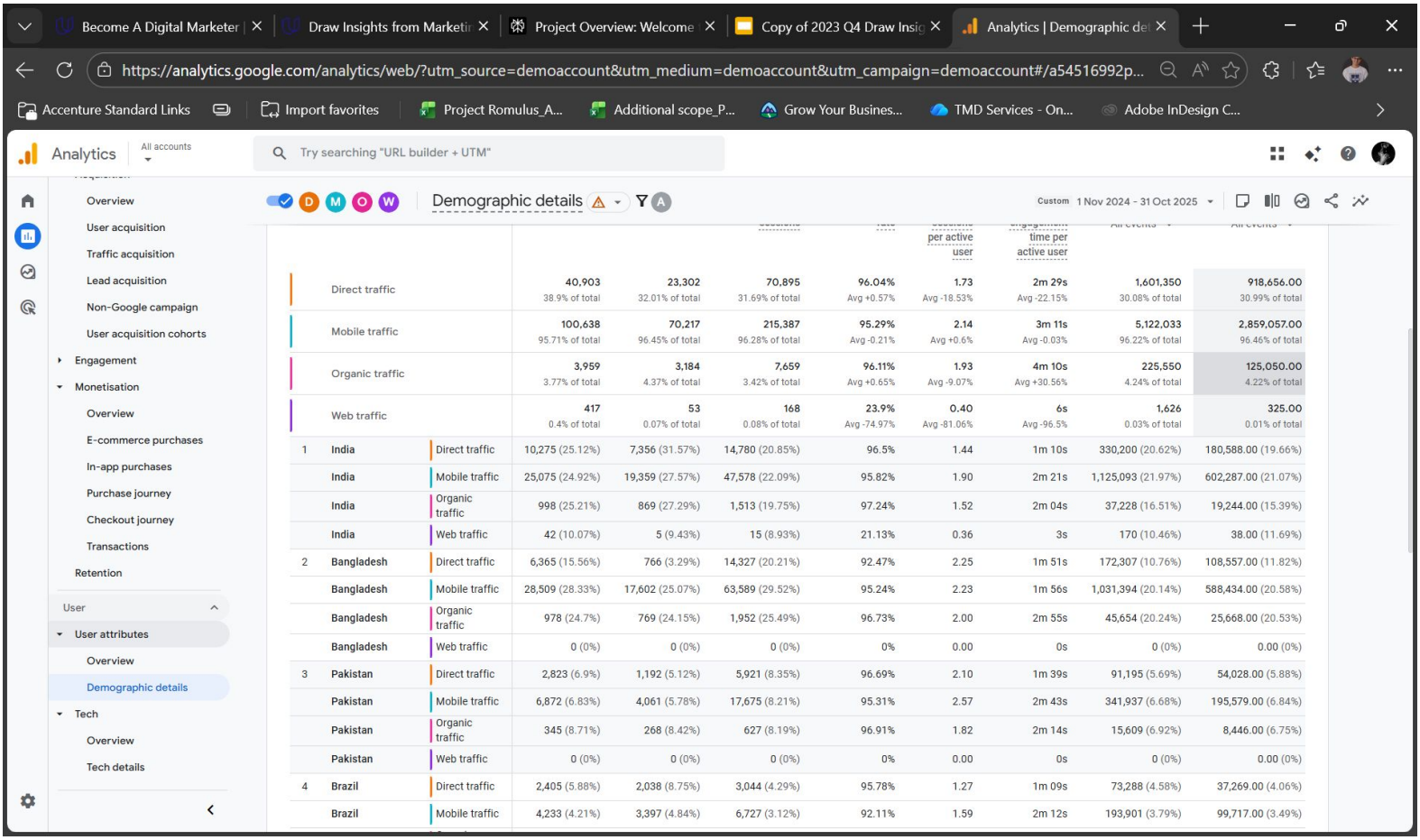
Ensure that the following are visible in the screenshot:

- Item names
- Number of items purchased
- Item revenue



Part Four: Segmentation

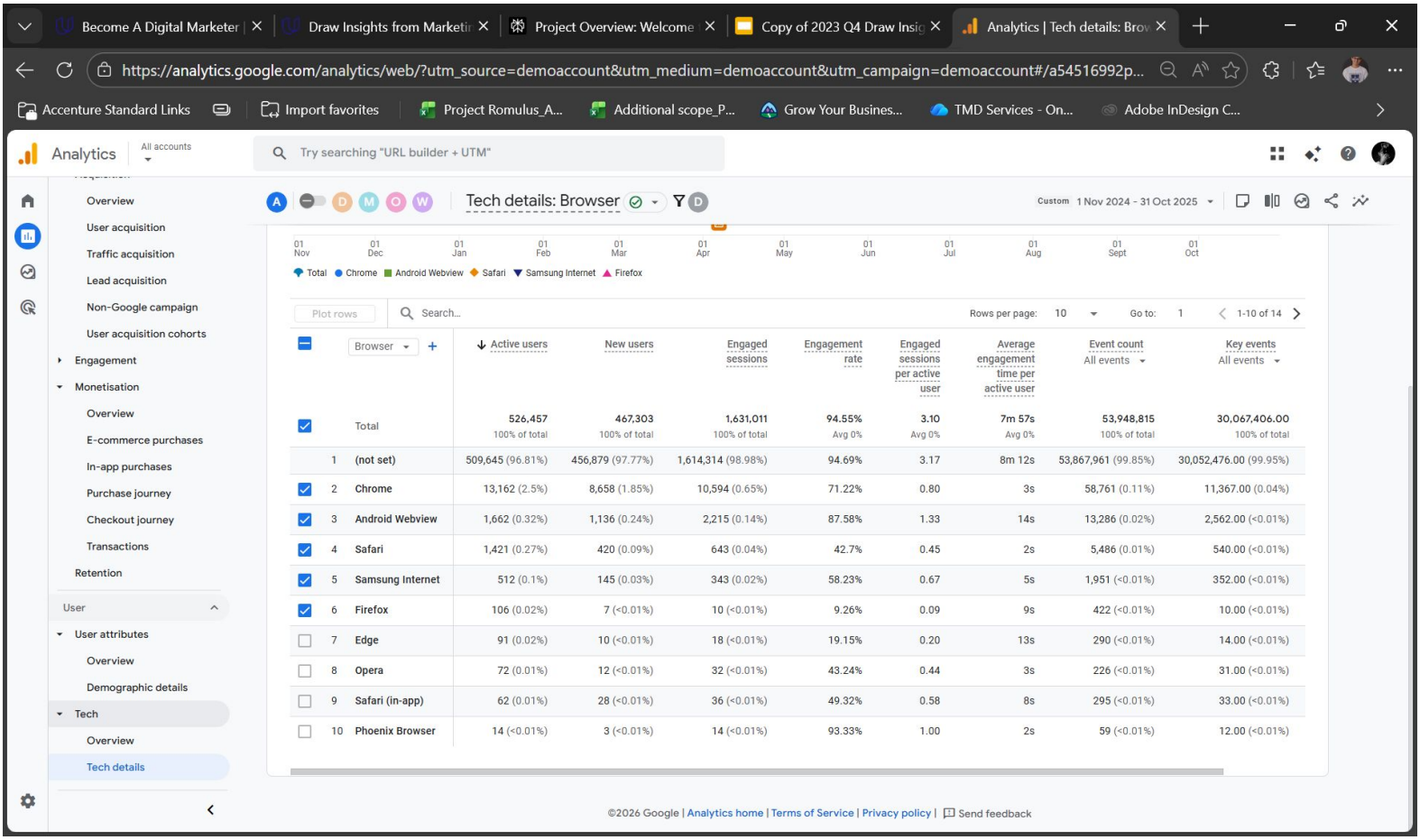
Audience Segment: Demographics



						per active user	time per active user		
	Direct traffic		40,903 38.9% of total	23,302 32.01% of total	70,895 31.69% of total	96.04% Avg +0.57%	1.73 Avg -18.53%	2m 29s Avg -22.15%	1,601,350 30.08% of total	918,656.00 30.99% of total
	Mobile traffic		100,638 95.71% of total	70,217 96.45% of total	215,387 96.28% of total	95.29% Avg -0.21%	2.14 Avg +0.6%	3m 11s Avg -0.03%	5,122,033 96.22% of total	2,859,057.00 96.46% of total
	Organic traffic		3,959 3.77% of total	3,184 4.37% of total	7,659 3.42% of total	96.11% Avg +0.65%	1.93 Avg -9.07%	4m 10s Avg +30.56%	225,550 4.24% of total	125,050.00 4.22% of total
	Web traffic		417 0.4% of total	53 0.07% of total	168 0.08% of total	23.9% Avg -74.97%	0.40 Avg -81.06%	6s Avg -96.5%	1,626 0.03% of total	325.00 0.01% of total
1	India	Direct traffic	10,275 (25.12%)	7,356 (31.57%)	14,780 (20.85%)	96.5%	1.44	1m 10s	330,200 (20.62%)	180,588.00 (19.66%)
	India	Mobile traffic	25,075 (24.92%)	19,359 (27.57%)	47,578 (22.09%)	95.82%	1.90	2m 21s	1,125,093 (21.97%)	602,287.00 (21.07%)
	India	Organic traffic	998 (25.21%)	869 (27.29%)	1,513 (19.75%)	97.24%	1.52	2m 04s	37,228 (16.51%)	19,244.00 (15.39%)
	India	Web traffic	42 (10.07%)	5 (9.43%)	15 (8.93%)	21.13%	0.36	3s	170 (10.46%)	38.00 (11.69%)
2	Bangladesh	Direct traffic	6,365 (15.56%)	766 (3.29%)	14,327 (20.21%)	92.47%	2.25	1m 51s	172,307 (10.76%)	108,557.00 (11.82%)
	Bangladesh	Mobile traffic	28,509 (28.33%)	17,602 (25.07%)	63,589 (29.52%)	95.24%	2.23	1m 56s	1,031,394 (20.14%)	588,434.00 (20.58%)
	Bangladesh	Organic traffic	978 (24.7%)	769 (24.15%)	1,952 (25.49%)	96.73%	2.00	2m 55s	45,654 (20.24%)	25,668.00 (20.53%)
	Bangladesh	Web traffic	0 (0%)	0 (0%)	0 (0%)	0%	0.00	0s	0 (0%)	0.00 (0%)
3	Pakistan	Direct traffic	2,823 (6.9%)	1,192 (5.12%)	5,921 (8.35%)	96.69%	2.10	1m 39s	91,195 (5.69%)	54,028.00 (5.88%)
	Pakistan	Mobile traffic	6,872 (6.83%)	4,061 (5.78%)	17,675 (8.21%)	95.31%	2.57	2m 43s	341,937 (6.68%)	195,579.00 (6.84%)
	Pakistan	Organic traffic	345 (8.71%)	268 (8.42%)	627 (8.19%)	96.91%	1.82	2m 14s	15,609 (6.92%)	8,446.00 (6.75%)
	Pakistan	Web traffic	0 (0%)	0 (0%)	0 (0%)	0%	0.00	0s	0 (0%)	0.00 (0%)
4	Brazil	Direct traffic	2,405 (5.88%)	2,038 (8.75%)	3,044 (4.29%)	95.78%	1.27	1m 09s	73,288 (4.58%)	37,269.00 (4.06%)
	Brazil	Mobile traffic	4,233 (4.21%)	3,397 (4.84%)	6,727 (3.12%)	92.11%	1.59	2m 12s	193,901 (3.79%)	99,717.00 (3.49%)

Values used:	*Age exactly matches 25-34*

Audience Segment: Technology



Browser	Active users	New users	Engaged sessions	Engagement rate	Engaged sessions per active user	Average engagement time per active user	Event count All events	Key events All events
Total	526,457 100% of total	467,303 100% of total	1,631,011 100% of total	94.55% Avg 0%	3.10 Avg 0%	7m 57s Avg 0%	53,948,815 100% of total	30,067,406.00 100% of total
1 (not set)	509,645 (96.81%)	456,879 (97.77%)	1,614,314 (98.98%)	94.69%	3.17	8m 12s	53,867,961 (99.85%)	30,052,476.00 (99.95%)
2 Chrome	13,162 (2.5%)	8,658 (1.85%)	10,594 (0.65%)	71.22%	0.80	3s	58,761 (0.11%)	11,367.00 (0.04%)
3 Android Webview	1,662 (0.32%)	1,136 (0.24%)	2,215 (0.14%)	87.58%	1.33	14s	13,286 (0.02%)	2,562.00 (<0.01%)
4 Safari	1,421 (0.27%)	420 (0.09%)	643 (0.04%)	42.7%	0.45	2s	5,486 (0.01%)	540.00 (<0.01%)
5 Samsung Internet	512 (0.1%)	145 (0.03%)	343 (0.02%)	58.23%	0.67	5s	1,951 (<0.01%)	352.00 (<0.01%)
6 Firefox	106 (0.02%)	7 (<0.01%)	10 (<0.01%)	9.26%	0.09	9s	422 (<0.01%)	10.00 (<0.01%)
7 Edge	91 (0.02%)	10 (<0.01%)	18 (<0.01%)	19.15%	0.20	13s	290 (<0.01%)	14.00 (<0.01%)
8 Opera	72 (0.01%)	12 (<0.01%)	32 (<0.01%)	43.24%	0.44	3s	226 (<0.01%)	31.00 (<0.01%)
9 Safari (in-app)	62 (0.01%)	28 (<0.01%)	36 (<0.01%)	49.32%	0.58	8s	295 (<0.01%)	33.00 (<0.01%)
10 Phoenix Browser	14 (<0.01%)	3 (<0.01%)	14 (<0.01%)	93.33%	1.00	2s	59 (<0.01%)	12.00 (<0.01%)

Values used:	*Write the values used to create the segment here*

Part Five:
Analysis and Suggestions

Google Merchandise Store data

You can find the results of the Google Merchandise Store campaigns below.

Campaign Name	Cost	Revenue	ROAS
Tech Trends: Discover the Latest Google Gear	$5,000	$3,000	0.6
Shop with Google: Unleash Your Digital Lifestyle	$5,000	$8,000	1.6
Google Gadgets Galore: Elevate Your Tech Game	$5,000	$8,000	1.6
Gear Up with Google: Your One-Stop Tech Shop	$8,000	$13,000	1.625
Google Merch Madness: Score Big on Tech Essentials	$5,000	$2,000	0.4
Unlock the Power of Google: Shop the Best in Tech	$2,000	$3,500	1.75
Totals	$30,000	$37,500	

Business Sales Growth

Based on the data provided, how might the campaigns be realigned and improved to potentially achieve a 20% YOY sales growth **without additional cost**? You can assume that the data will remain consistent over the projected time frame. Please reference specific data to support your answer, such as metrics and campaigns.

You could get the answer by asking yourself: Which campaign would I spend less, and which would I spend more?

To achieve 20% YoY revenue growth ($37,500 → $45,000 = +$7,500) without additional cost:

Reduce spend on low-ROAS campaigns:
-"Google Merch Madness" (ROAS 0.4): Cut $5,000 → $1,000 (save $4,000)
-"Tech Trends" (ROAS 0.6): Cut $5,000 → $1,500 (save $3,500)

Reallocate $7,500 to high-ROAS campaigns:
-"Unlock the Power of Google" (ROAS 1.75): +$5,000 = +$8,750 revenue
-"Gear Up with Google" (ROAS 1.625): +$2,500 = +$4,063 revenue

Total projected revenue: $37,500 + $12,813 = $50,313 (34% growth > 20% target). Uses campaign-specific ROAS data, assumes constant performance.

eCommerce improvements

Looking at your website pages or the Google Merchandise Store website and current eCommerce experience, identify one change to the eCommerce UX and one additional eCommerce option you would recommend implementing.

Example: One way to improve eCommerce capabilities would be to add the option of a digital wallet with the option to securely store and manage cards that have been used for payment, along with the option of using PayPal or Apple Pay.

UX change: Add persistent mini-cart preview on mobile checkout (52% traffic). Shows items/price/total instantly, reducing 20% abandonment from navigation confusion. Test via A/B on Google Merch store.

Other eCommerce change or addition: Implement one-click Google Pay/Apple Pay for returning users. Saves card details securely, boosting conversions 2x (from 1.5% baseline). Complements current cart, targets high mobile abandonment.

Technology

It is time for some exploration! You need to find 2 emerging marketing technologies that you could use in a technology stack. For each one, you need to describe why you would use that tool.	
1	*Agentic AI (autonomous agents)*
	Automates bid optimization/creative testing across Google Ads/YouTube, improving ROAS 30%+ by self-adjusting in real-time—perfect for MCN scale
2	*Technology Generative AI video personalization*
	Creates custom product demo videos per user (e.g., "Android hoodie for creators"), boosting engagement 25% via dynamic thumbnails/CTAs